Last name _____

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LARSON—MATH 756—CLASSROOM WORKSHEET 17
Knuth—Sandwich Theorem

Review

1. What linear algebra results is Knuth referring to?
2. What is the *characteristic labeling* of a set $S \subseteq V(G)$?
3. What is a *stable labeling* of a stable set $S \subseteq V(G)$?
4. What is a *clique labeling* of a clique $S \subseteq V(G)$?
5. What is $\text{STAB}(G)$ for a graph G?
6. Find $\text{STAB}(G)$ for the paw graph.
7. What is $\text{QSTAB}(G)$ for a graph G?
8. Why is $\text{QSTAB}(G)$ convex?

New

1. Why is $\text{STAB}(G) \subseteq \text{QSTAB}(G)$?

2. What is the *cost* $c(\hat{a}_v)$ of a (unit) vector $\hat{a}_v$ (for $v \in V$) in an orthonormal labeling of a graph G?

3. Show: for a graph G, any independent set S, and any orthonormal labeling $\{a_v\}$, $\sum_{v \in S} c(\hat{a}_v) \leq 1$.

4. What is TH(G) for a graph G?

5. Find TH(G) for the paw graph.

6. Why is TH(G) "obviously" convex?

7. Show: STAB(G) $\subseteq TH(G)$, for any graph G.

8. What is Knuth's first definition of ϑ?

9. What is Knuth's proof that $\alpha \leq \vartheta$?